PROJECT ABSTRACT

Design and Verification of a RISC-V VeeR EL2 Based Communication SoC with DMA-Accelerated SPI and I²C

1. Problem Statement

Embedded systems that communicate over serial protocols such as SPI and I²C often move data one byte or one word at a time, with the processor core issuing a register access and servicing an interrupt for every transfer unit. For high-throughput or continuous data-acquisition applications, this CPU-driven, byte-by-byte servicing model consumes a significant share of the core's execution cycles and limits the achievable transfer rate, even though the actual data movement between a peripheral register and memory does not require program logic.

This project proposes the design and verification of an RTL-based Communication SoC centered around the faculty-provided VeeR EL2 processor core, an RV32IMC RISC-V implementation from the CHIPS Alliance. The proposed SoC integrates an SPI Controller, an I²C Controller, and a DMA Controller so that block data transfers to and from external SPI and I²C devices can be carried out with minimal core intervention. The VeeR EL2 core will configure the peripherals and set up DMA transfer descriptors, after which the DMA Controller autonomously moves data between the peripheral FIFOs and the core's Data Closely-Coupled Memory (DCCM), notifying the core only on completion or error. UART will be used to report transfer status and results to a host computer.

The main problem addressed by the project is offloading repetitive, per-unit data movement from the RISC-V core to dedicated hardware, while preserving a simple, memory-mapped software interface and remaining consistent with the access rules that the VeeR EL2 core imposes on its dedicated DMA slave port. The project will demonstrate how a system-level DMA Controller can access a RISC-V core's closely-coupled memory directly, in parallel with core execution, to accelerate SPI and I²C communication.

2. Proposed SoC Architecture

The proposed SoC will use the faculty-provided VeeR EL2 RISC-V RTL processor core as the central processing unit. Per the VeeR EL2 Programmer's Reference Manual, the core exposes a dedicated DMA slave port, separate from its IFU and LSU bus master interfaces, intended for read/write access to the core's ICCM and DCCM by external bus masters such as an on-chip DMA controller. This project uses that port as the data path for bulk SPI/I²C transfers: the CPU configures the SPI Controller, I²C Controller, and DMA Controller over the AHB-Lite system bus (via the LSU interface), while the DMA Controller itself connects to VeeR EL2's DMA slave port to write received data into, or read data to transmit from, the DCCM directly.

Two access-ordering rules from the manual shape the design. First, DMA write accesses to the ICCM/DCCM must be 32- or 64-bit aligned and sized, since ECC is generated per word/double-word and unaligned partial writes would require a read-modify-write the DMA port does not support. Second, the core's own DCCM accesses take priority over the DMA slave port whenever the DMA FIFO is not full, so the DMA Controller's transfer rate can be affected by concurrent core activity; the mfdc register's dqc field governs how long a blocked DMA access waits before requesting a pipeline bubble. The design and testbench will respect both rules, and because the core and the DMA slave port give no ordering guarantee on accesses to the same address, the software will only read a DCCM buffer after the DMA Controller signals completion via the PIC.

Block	Primary Function
VeeR EL2 Processor Core	RV32IMC RISC-V core (CHIPS Alliance VeeR EL2) that executes the embedded application, configures peripherals and DMA transfer descriptors over its LSU system bus master interface, and services completion interrupts.
ICCM	Instruction Closely-Coupled Memory: core-local, ECC-protected memory that stores instructions, accessed directly by the IFU.
DCCM	Data Closely-Coupled Memory: core-local, ECC-protected memory that stores program data and, in this project, the source/destination buffers for DMA-driven SPI and I²C transfers.
System Bus Interconnect (AHB-Lite)	Routes CPU-initiated LSU read/write transactions to the memory-mapped control/status registers of the SPI Controller, I²C Controller, DMA Controller, and UART.
DMA Controller	External bus master that autonomously moves data between the SPI/I²C peripheral FIFOs and the DCCM through VeeR EL2's dedicated DMA slave port, without CPU intervention on a per-byte basis.
PIC	VeeR EL2's built-in Programmable/Platform-Level Interrupt Controller. Aggregates DMA-complete, SPI, and I²C interrupt requests (up to 255 sources) and presents prioritized interrupts to the core.
UART	Provides communication between the SoC and a host computer for reporting transfer status and results.
SPI Controller	Provides synchronous serial communication with an external SPI-compatible device; its transmit/receive data registers act as a DMA source or destination.
I²C Controller	Provides two-wire serial communication with I²C-compatible sensors or peripheral devices; its data register acts as a DMA source or destination.

Functional architecture: The VeeR EL2 core communicates with the SPI Controller, I²C Controller, and DMA Controller through the AHB-Lite bus interconnect using memory-mapped control and status registers, all located in a non-cacheable, side-effect-marked address region distinct from the core-local ICCM/DCCM region. To start a transfer, the core writes a descriptor (source, destination, length, direction) to the DMA Controller and enables the corresponding peripheral; the DMA Controller then moves data between the peripheral's FIFO and the DCCM over the DMA slave port without further CPU involvement, and raises a PIC interrupt on completion or error. UART provides a convenient interface for observing transfer status and data received from external devices.

3. Selected Additional IPs

1. SPI Controller: The SPI Controller will provide a synchronous serial interface for communicating with an external SPI-compatible device, managing serial clock, chip-select, and transmit/receive data. Its data FIFO will be configurable as a DMA source (receive) or destination (transmit), and it will raise a PIC-routed interrupt on FIFO threshold and transfer completion.

2. I²C Controller: The I²C Controller will provide a two-wire serial interface for communicating with I²C-compatible sensors or peripheral devices, handling start, address, read/write, data transfer,

acknowledgement, and stop conditions. Like the SPI Controller, its data register will be a DMA source or destination, with status and interrupt signaling to the core through its registers and PIC connection.

3. DMA Controller: The DMA Controller will be a memory-mapped peripheral, configured by the core over the AHB-Lite bus with a transfer descriptor (peripheral select, direction, DCCM address, and length). Once triggered, it acts as an external bus master on VeeR EL2's DMA slave port, issuing 32- or 64-bit aligned reads or writes to the DCCM as it moves data to or from the selected peripheral's FIFO, and signals the core via the PIC when the transfer completes or a bus error occurs.

4. Expected System Operation and Verification

The embedded application running on the VeeR EL2 core will initialize the DCCM buffers and peripherals, configure the SPI and I²C interfaces, program DMA transfer descriptors, and enable the PIC's interrupt sources. The DMA Controller will then carry out block transfers between the SPI/I²C peripherals and the DCCM autonomously, while the core is free to execute other application code, and will interrupt the core on completion or error. UART will be used to transmit selected data and transfer status to a host computer for observation.

The complete SoC will be verified through RTL simulation using a dedicated testbench. Verification will include CPU-to-peripheral AHB-Lite bus transactions and address decoding, SPI transmission and reception, I²C data transfer, correct DMA slave port access sizing and alignment to the DCCM, DMA arbitration behavior against concurrent core accesses, PIC interrupt generation for DMA and peripheral events, UART communication, and overall application execution with and without CPU load concurrent with active DMA transfers. The project will demonstrate the integration of the VeeR EL2 RISC-V core, its DMA slave port and core-local memory, and DMA-accelerated serial communication peripherals in a complete RTL-based SoC.

Reference: RISC-V VeeR EL2 Programmer's Reference Manual, Rev. 1.4, CHIPS Alliance, Dec. 2022 (Section 2.14, DMA Slave Port).